DeepFake Video Detection System

A Robust Hybrid CNN-RNN Architecture featuring Temporal Attention Pooling for Synthetic Video Classification

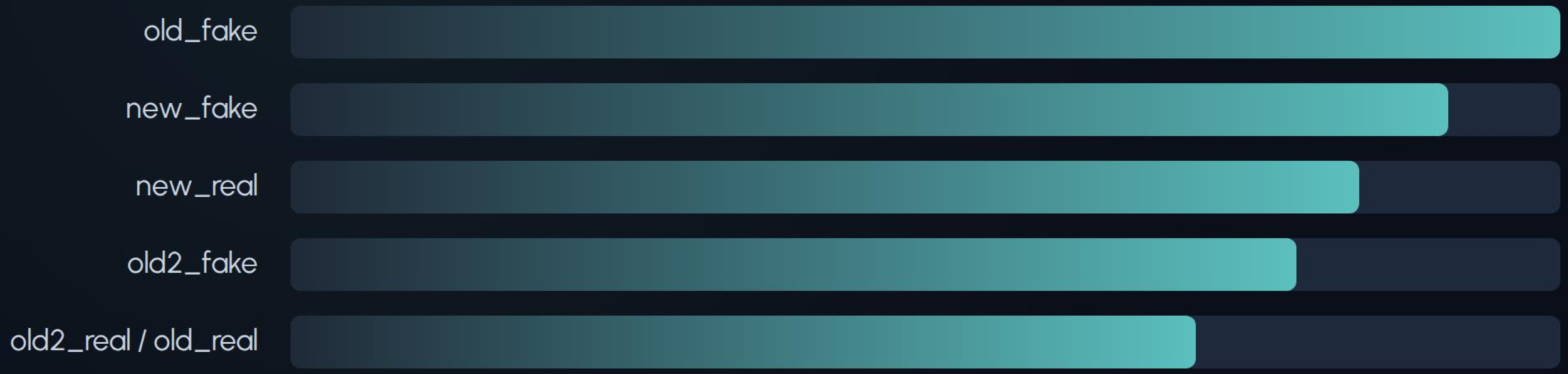
Project Overview

Harnessing spatial feature extraction and sequence modeling to counter deep learning-based video manipulations.

Dataset Characterization

- **Unique Structured Frame-Folders:** Every video clip is organized as an individual sequence of extracted image frames.
- **Dataset Balance:** Structured split featuring 1,888 synthetic (Fake) samples and 1,583 authentic (Real) samples.
- **Filtering Constraints:** Ensures validation validity by omitting files failing to meet the 32 frame minimum limit.
- **Source-Bias Control:** Cross-checked partitions (Train, Val, Test) to bypass specific source signature overfitting.

Source Data Volumes



Detailed source categories from 3,471 total processed video files across 6 distinguished fake and real datasets.

Pipeline Architecture

A step-by-step modular breakdown of video frames preprocessing, feature extraction, and temporal analysis.

Video Preprocessing Pipeline

Face Detection

Fast face identification utilizing MediaPipe framework, Haar Cascades, or OpenCV Caffe-SSD DNN models.

Dynamic Crop

Auto-crops the largest face. Extends bounding box coordinates with a fallback back to the original full image frame.

Data Augmentation

Applies standard color jitter, blur effects, Gaussian noise, and random JPEG compression for real-world robustness.

Core Processing Networks

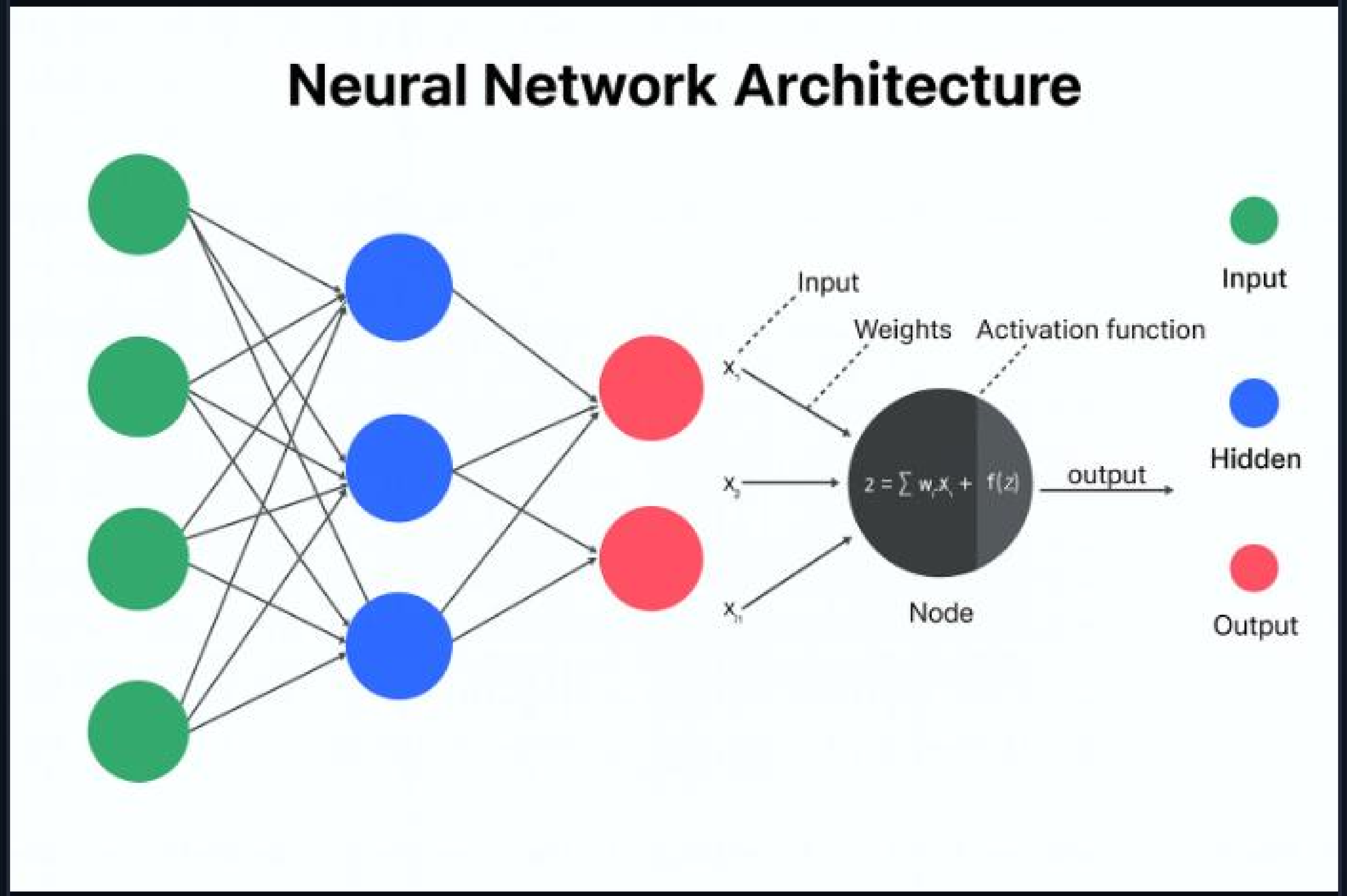
ResNeXt50 Feature Extractor

Pre-trained deep convolutional network Backbone used for high-fidelity spatial feature extraction. Features 2,048 flattened dimensions per visual frame, creating robust spatial representations.

BiLSTM Temporal Encoder

A bidirectional Long Short-Term Memory recurrent block. Connects chronological frame information in both directions with hidden layers to analyze subtle temporal editing traces.

Attention-Based Pooling



Temporal Attention Pooling

Calculates dynamic weights for each frame outputted from the BiLSTM block. Replaces traditional average pooling by selectively highlighting the most crucial synthetic transitions or artifact anomalies across the video sequence timeline.

Training Strategy

Configuring optimization profiles, model checkpoint routines, and early-stopping parameters.

Training Optimization

- **Optimization Setup:** Configured using AdamW optimizer with a base learning rate of 3e-5 and weight decay of 1e-4.
- **Mixed Precision Support:** Leverages PyTorch GradScaler for optimal GPU memory footprint and faster computation.
- **Smoothing Penalty:** Smooths cross-entropy losses (value 0.02) to deter excessive output prediction confidence.
- **Convergence Control:** Built-in dynamic ReduceLROnPlateau learning-rate scheduler alongside an early stopping threshold.

Evaluation & Performance

Quantifying prediction accuracies and identifying error thresholds on hold-out testing datasets.

Overall Test Accuracy

Correct Classification (96.16%)

Incorrect Classification (3.84%)

Validation-selected threshold of 0.30 evaluated across 521 unseen test partition videos.

Evaluation Metrics Dashboard

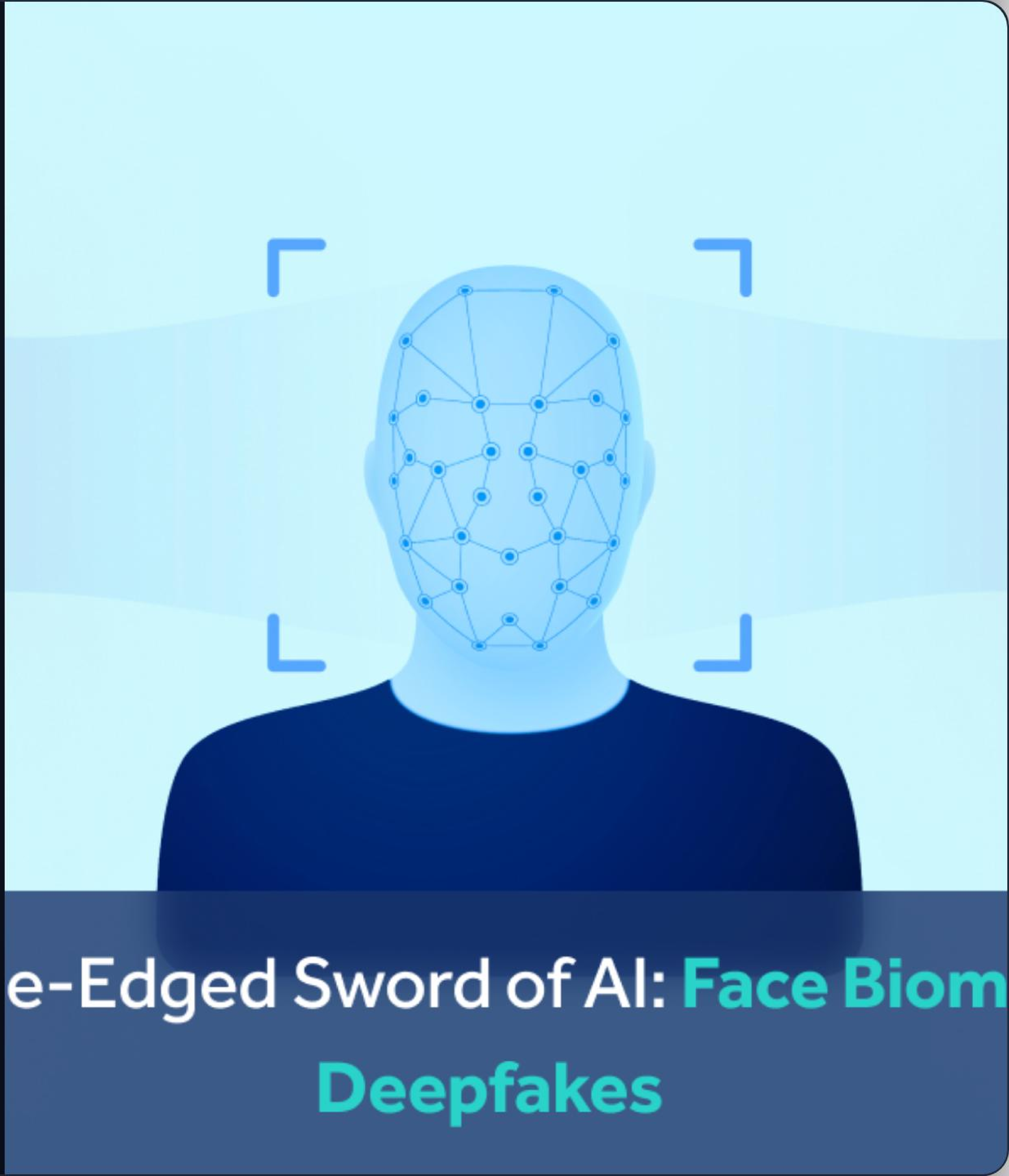
Video Target Class	Precision	Recall	F1-Score	Test Support
Fake (Synthetic)	0.96	0.96	0.96	284 clips
Real (Original)	0.96	0.96	0.96	237 clips
System Average (Macro)	0.96	0.96	0.96	521 clips

Inference Upgrades

Multi-Clip Support

Analyzes and partitions incoming raw videos dynamically, running inference across multiple segments for enhanced stability.

Offers automated face cropping utilizing OpenCV Haar Cascades, MediaPipe, or SSD modules, with adaptive full-frame center cropping as a fallback.



Questions & Answers

Thank you for your attention.

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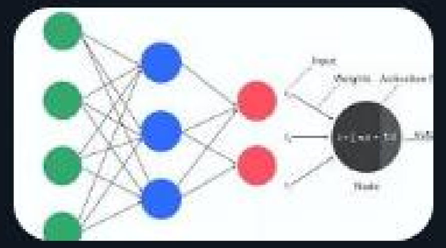
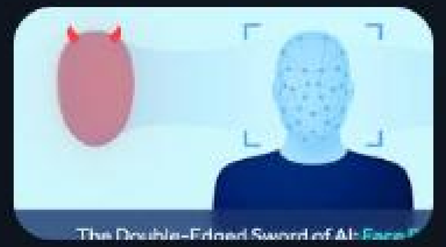
Image Sources



https://ik.imagekit.io/upgrad1/abroad-images/imageCompo/images/ChatGPT_Image_Nov_20_2025_03_47_56_PMTT18DE.png

Source: www.upgrad.com



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Source: facia.ai